

Wrong PSF Inflates Little Red Dot Hosts

Field-star PSFs misassign light when source spectra differ, faking extended hosts in compact high-redshift nuclei.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for little red dots and compact nuclei.

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Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from nearby field stars. This assumes the target's spectrum matches those stars across the broad filter.

No test existed for whether spectrally mismatched field-star PSFs bias host measurements of compact distant sources.

What this paper is selling

Broad-band stellar PSFs systematically misassign light for sources with different spectral slopes, overestimating or inventing extended hosts. A source-conditioned polychromatic PSF correction is derived and demonstrated for other compact distant sources.

Key results to remember

- A pure point source yields spurious Sérsic host flux in every broad filter.
- Every tested LRD spectrum, stellar reference, and detector position exceeds the published 11.8% host.
- Inferred sizes, luminosities, and masses of little red dots are therefore biased.

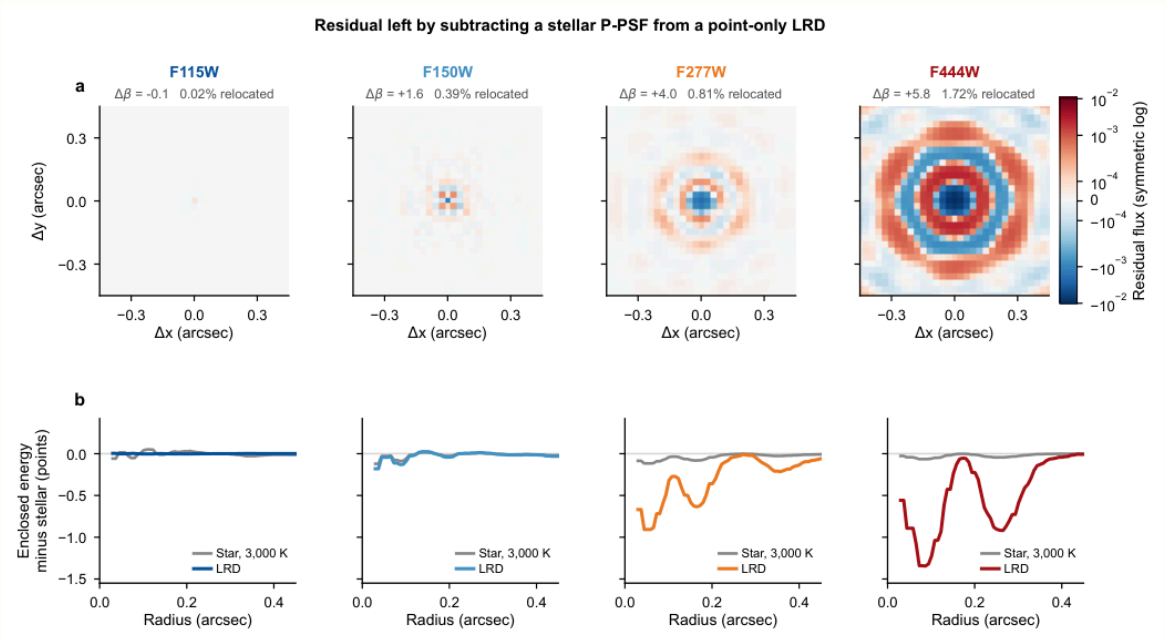


Fig. 1. Zero-host LRD input still leaves wavelength-dependent residuals, mimicking a Sérsic host component...

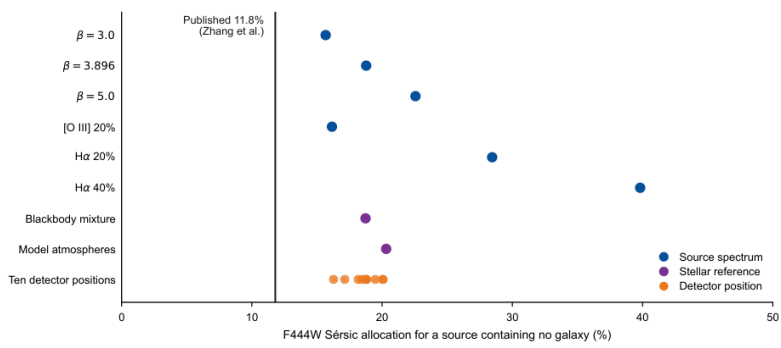


Fig. 2. All tested spectra, stellar references, and detector positions allocate more F444W flux...